Following the Fashion?

How ML research trends rise, who drives them, and whether they pay off.
A pre-registered study of 43,480 ICLR submissions, 2017–2026

[Your name]

[Venue / seminar], [date]

Machine learning moves in waves

papers proposing a new architecture

24% → **8%**

2017 2026

papers building on a foundation model

0.3% → **54%**

2017 2026

titles branded "NAME: subtitle"

9% → **41%**

2017 2026

Are these fashions? Who starts them, who follows, and does following them pay off?

Three questions, three measurement traps

Questions

RQ1 Rise What does a rising fashion crowd out?

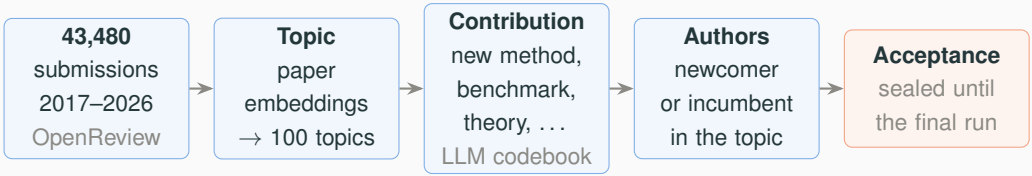
RQ2 Follow Who brings new kinds of work: newcomers or incumbents?

RQ3 Reward Does following the fashion get papers accepted?

Traps, and our answer

- **Survivorship**: usually only accepted papers are visible
  → all submissions: accepted, rejected, withdrawn
- **Hindsight**: topics defined with today's eyes
  → prospective topic maps
- **Forking paths**: many ways to analyse
  → pre-registration, one frozen run

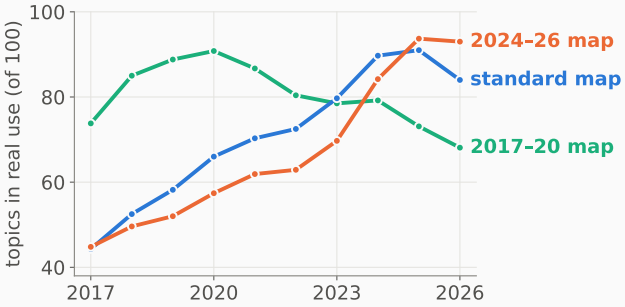
Data: ICLR submissions, accepted or not



- Topics match ICLR's own areas better than a keyword baseline
- Labels: a blind second labelling pass agrees, $\kappa = 0.93$

- **Newcomer paper**: no author has submitted to ICLR *in this topic* before (73%)
- All comparisons are **within the same topic and year**

Contribution 1: topic maps carry hindsight



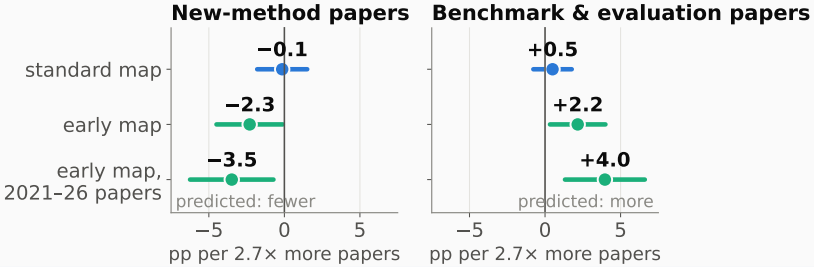
- 69% of papers are from 2024–26, so the standard map, fitted on all years, is **a map of today**
- “Is ML getting more diverse?” Each map answers: most diverse in the years it was built from

The topic map is not neutral preprocessing: it decides which past you see.

. . . and hindsight changes the conclusions

"Emerging" topics ($\geq 3\times$ growth): **8** with the 2017–20 map vs. **29** with the standard map

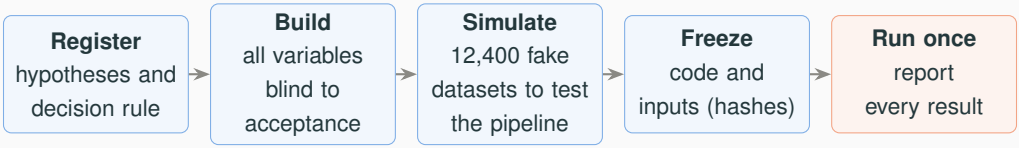
Do topics age, from new methods towards benchmarks?



Same papers, same model, different map: "no lifecycle" becomes "lifecycle".

So every trend result must also hold on prospective maps.

Contribution 2: a design that keeps us from fooling ourselves



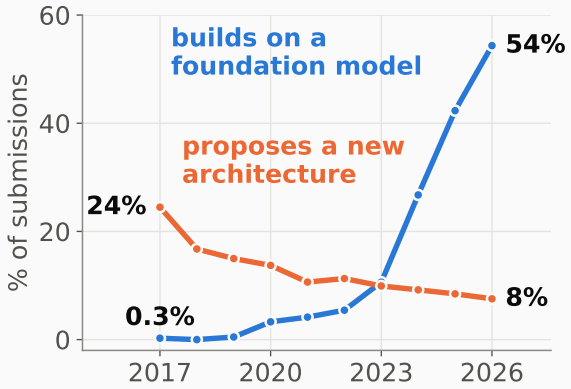
A claim needs all four:

1. the predicted direction
2. corrected $p < 0.05$ (Holm)
3. the same direction in $\geq$75% of 10 alternative analyses
4. agreement on the prospective maps

The simulation caught a real problem

In a realistic no-effect simulation, standard clustered errors gave 12.6% false positives where 5% was promised. We report two extra checks for every key result.

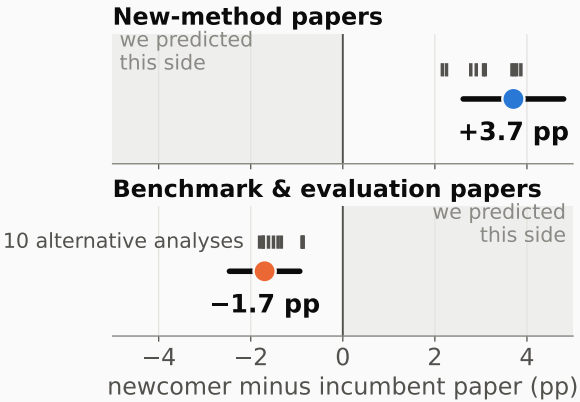
Finding 1: foundation models crowd out new architectures



Within the same topic: the more of last year's papers built on foundation models, the fewer new architectures this year.

- 10% → 50% FM use: about one third fewer architecture papers
- 0 → 100%: −8 pp (95% CI −12 to −4), on a 9% base
- Pre-registered, **supported**; holds in all 9 alternative analyses

Finding 2: newcomers bring new methods, incumbents build benchmarks

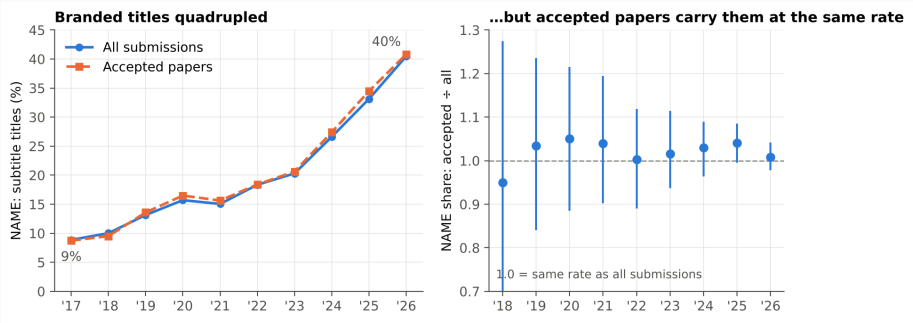


We predicted the opposite: newcomers follow, incumbents innovate.

- Newcomer papers: more new methods, a fifth fewer benchmark and evaluation papers
- Same direction in all 10 analyses and both extra checks

Same topic and year. The benchmark gap is partly team size (−0.5 pp with team size controlled); the method gap is not.

Finding 3: fashion in style, too



Branded titles quadrupled, yet accepted papers carry them at the same rate as all submissions (ratio 1.02).

Not just LLM writing tools: branding rises in papers with and without LLM-style wording.

Exploratory analysis, not part of the pre-registration.

Does following the fashion pay off? We find no sign that it does

Crowded topics

No acceptance premium; if anything lower.

-1.7 pp per $2.7\times$ more prior papers, not significant

Branded titles

No premium.

accepted papers use them at the submitted rate

Against the tide

Where foundation models dominate, the remaining architecture papers do relatively better.

$+8$ pp, exploratory

Fashion spreads, but the review process does not visibly reward following it.

What does go with acceptance (associations, not advice): theory papers ($+9$ pp vs. new methods), larger teams. See backup.

Contributions

1. **A measurement warning, with a fix.** Topic maps fitted in hindsight invent "emerging" topics and can flip trend conclusions. Prospective maps are a cheap check.
2. **A rigorous template for science-of-science.** Pre-registered, outcome-blind, simulation-checked, one frozen run.
3. **New evidence on research fashions.**
   - foundation models crowd out new architectures
   - newcomers to a topic, not incumbents, bring new methods
   - style spreads like topics, without a reward

Limits: one venue; associations, not causes; labels from an LLM ($\kappa = 0.93$, not yet checked by humans); author IDs changed in 2021, which undercounts incumbents and makes the newcomer results conservative.

Next steps

- **Replication on held-out ICLR papers**: the four key hypotheses are already registered
- **Other venues**: ACL Anthology / ARR
- **Human check** of a sample of contribution labels
- **Review scores**, to separate reviewers from final decisions

Thank you. Questions?

Backup

B1. All pre-registered hypotheses

	Prediction	Estimate (95% CI)	Verdict
H1a	fewer new methods in mature topics	−0.1 pp (−1.8, +1.5)	not supported
H1b	more benchmarks in mature topics	+0.5 pp (−0.8, +1.8)	not supported
H1c	more empirical studies in mature topics	−0.3 pp (−1.3, +0.7)	not supported
H1d	**fewer new architectures after FM adoption**	**−8.0** pp (−12.1, −3.8)	**supported**
H1e	more empirical/evaluation after FM adoption	−0.6 pp (−3.6, +2.4)	not supported
H2a	fewer incumbents in hot topics	−0.7 pp (−2.7, +1.2)	inconclusive
H2b	**newcomers: fewer new methods**	**+3.7** pp (+2.6, +4.8)	**significant, opposite**
H2c	**newcomers: more benchmarks**	**−1.7** pp (−2.5, −0.9)	**significant, opposite**
H3a	lower acceptance in mature topics	−1.7 pp (−3.7, +0.3)	inconclusive (low power)
H3b	… more so for method papers	−0.4 pp (−1.5, +0.8)	inconclusive

Maturity effects are per log unit of prior papers in the topic (2.7× more). H2d/H3c (buzzwords) were dropped before any estimate because the measure failed its blind validation.

B2. Newcomer results: robustness

	New methods	Benchmarks & eval.
Main estimate	$+3.7$ pp	-1.7 pp
Range over 10 alternative analyses (all $p < 0.05$)	$+2.2$ to $+3.9$	-1.8 to -0.9
Two-way clustering (topic $\times$ early-map topic)	$+3.7$	-1.7
Team size controlled	$+3.9$	-0.5 ($p = 0.07$)
Only years with consistent author IDs	$+3.7$	-1.7
Ignoring the paper's own earlier versions	$+3.8$	-1.7
Corrected for label errors	$+4.1$	-1.8

Base rates: 75% of papers are new methods, 7.7% benchmark or evaluation; 73% are newcomer papers. Confirmatory p from a wild cluster bootstrap: $p < 0.001$ for both. By era, the method gap appears in every era (2019–20: $+5.5$ pp).

B3. Why we did not trust standard errors

- Before seeing any result, we simulated 12,400 datasets with **no true effect** but realistic topic structure.
- When the true topic structure cuts across our 100 topics, the standard test rejected 12.6% (method) and 10.0% (benchmark) of the time, not 5%. A wild bootstrap did not fix it.
- So we added, before the run:
  - two-way clustering (two different topic maps at once);
  - a stress test: shrink each t-statistic by the inflation seen in simulation (1.31 / 1.22).
- Newcomer results: $t = 6.7$ and -4.4. After the stress test, Holm $p = 4 \times 10^{-6}$ and 10^{-3}.

Lesson: check your test on data shaped like yours before trusting its p-values.

B4. Are the topics and labels any good?

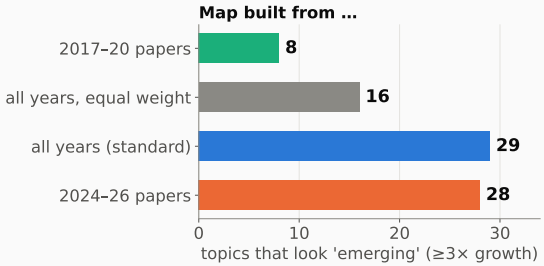
Topics (SPECTER2, 100 clusters)

- Agreement with ICLR's own primary areas (2024–26): NMI 0.30 vs. 0.27 for a keyword (TF-IDF) baseline
- Blind naming test: from title and abstract, a checker picks the right topic name 61% of the time (chance 10%)
- Stable across refits (ARI 0.66)

Contribution labels (LLM, fixed codebook)

- Blind second pass on 2,174 papers: $\kappa = 0.93$
- 95.8% agreement with 2,300 reference labels (also LLM-made, so this is consistency, not human validity)
- A correction for label errors makes the newcomer results slightly larger

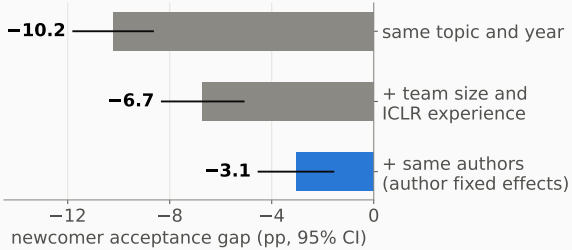
B5. Hindsight: more numbers



- Share of 2024–26 papers in “emerging” topics: 22% with the early map vs. 41% with the standard map
- “Declining” topics: 13 with the standard map vs. 26 with the year-balanced map, so faded topics get absorbed into today’s topics
- Fit of papers to their topic: standard map 0.62 (2017) vs. 0.65 (2026); early map 0.69 vs. 0.53

All maps have 100 topics; only the years used to build them differ.

B6. Are newcomer papers accepted less often? (exploratory)

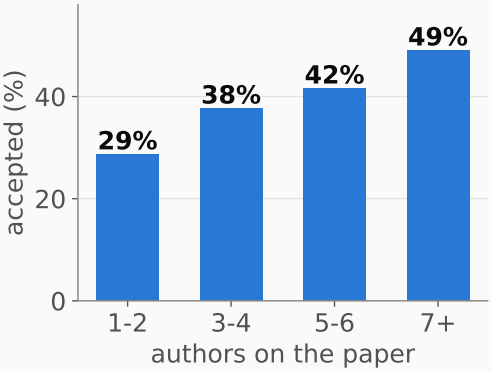


- Raw: 37% vs. 47% accepted
- The gap shrinks as we compare more similar papers; within the same authors it is about −3 pp
- The same person entering a topic new to them: −1.4 pp

Mostly about *who* writes newcomer papers. ICLR is double-blind, so the rest need not be bias: being new to a topic can make a paper weaker.

Author-FE sample: −5.4 pp without, −3.1 pp with author fixed effects.

B7. Team size and acceptance (exploratory)



- +8.8 pp per doubling of team size, same topic and year
- Steepest for benchmark papers (1–2 authors 21% vs. 7+ authors 52%)
- Flattest for theory (1–2 authors 42%)
- Positive in all 20 areas; flattest in federated learning and time series

Team size also stands for compute, data and seniority. Not advice to add authors.

B8. Acceptance by contribution type (exploratory)

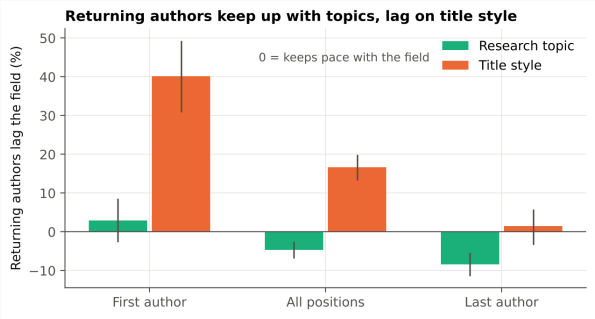
Same topic and year, relative to new-method papers:

Theory	$+8.9$ pp
Dataset / benchmark	$+3.9$ pp
Evaluation	-1.9 pp (n.s.)
Empirical study	-2.7 pp
Other (systems, applications, position papers, . . .)	-8.0 pp

Papers building on a foundation model: $+4.6$ pp vs. papers that do not.

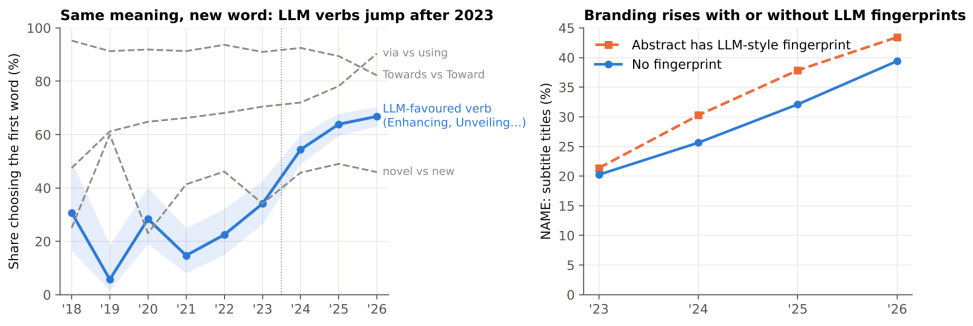
Unadjusted, exploratory. Selection is likely: who writes theory papers differs from who writes method papers.

B9. Title style: who carries it?



- Returning authors move into new topics as fast as the field
- But they lag on the new title style, most as first authors
- *Topics change with the field; style changes with the people*

B10. Is it just LLM writing tools?



Same meaning, new word: LLM-favoured words (“Enhancing”, “Unveiling”) jump after 2023 (odds ratio 2.2), while control word pairs do not.

But the “NAME: subtitle” structure rises with or without LLM fingerprints in the abstract.

B11. Definitions

Topic	one of 100 clusters of paper embeddings (title + abstract)
Topic maturity	$\log(1 + \text{earlier papers in the topic})$; $+1$ = 2.7$\times$ more papers
Hot topic	top fifth of topics by last year's growth
Newcomer paper	no author has an earlier ICLR submission (any outcome) in that topic
FM adoption	share of the topic's papers last year that built on a foundation model
Benchmark & eval.	primary contribution is a dataset, benchmark or evaluation
Model	linear probability model with topic and year fixed effects; errors clustered by topic
pp	percentage points